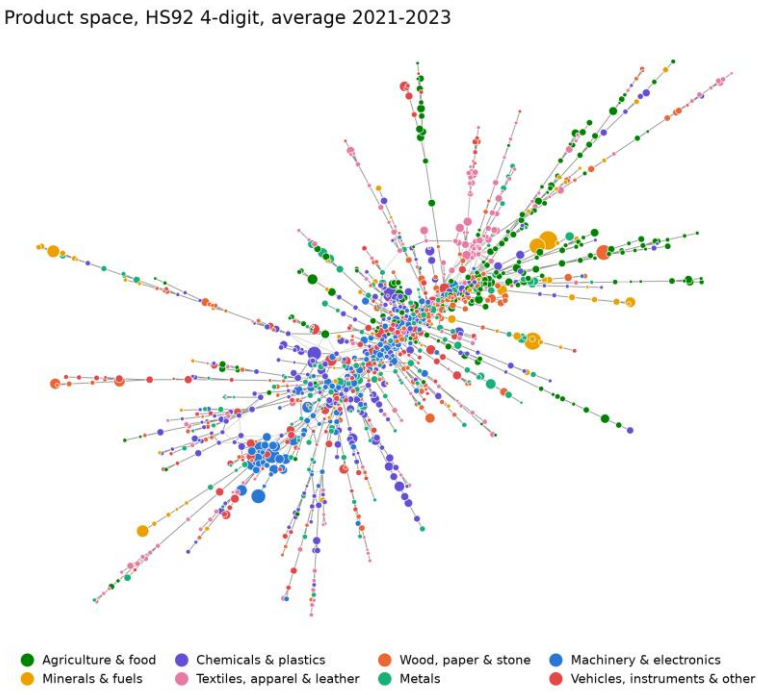
RELATEDNESS NETWORK FROM OPEN DATA: THE PRODUCT SPACE

RÉKA BALÁZS – EXTRA HOMEWORK 2

1. Data and method

I used the Atlas of Economic Complexity trade data from the Harvard Growth Lab (file hs92_country_product_year_4.csv). It shows how much each country exports of each 4-digit product from 2000 to 2024. I took the average of 2021–2023 and kept countries that export at least $1 billion a year (159 countries) and products with at least $10 million in world exports (1,230 products).

I followed Hidalgo et al. (2007). A country counts as an exporter of a product if its RCA is at least 1. Two products are close if the same countries export both. Proximity is the number of countries exporting both, divided by the number exporting the more common one. To draw the network, I kept the maximum spanning tree and added the strongest links until each product had 4 links on average (all added links have proximity ≥ 0.545). Colors show 8 sectors grouped from the HS chapters, and node size shows world exports. Gold and other precious metals are in the "Wood, paper & stone" group.



1. Figure

The picture looks like the product space in Hidalgo et al. (2007). Machinery, electronics, vehicles and chemicals form a dense area in the middle and lower left. Garments make a tight

pink cluster at the top. Agricultural products, minerals and fuels sit on long branches at the edge. Crude oil, refined oil and natural gas, three of the five largest products, sit at the edge with only one or two links each.

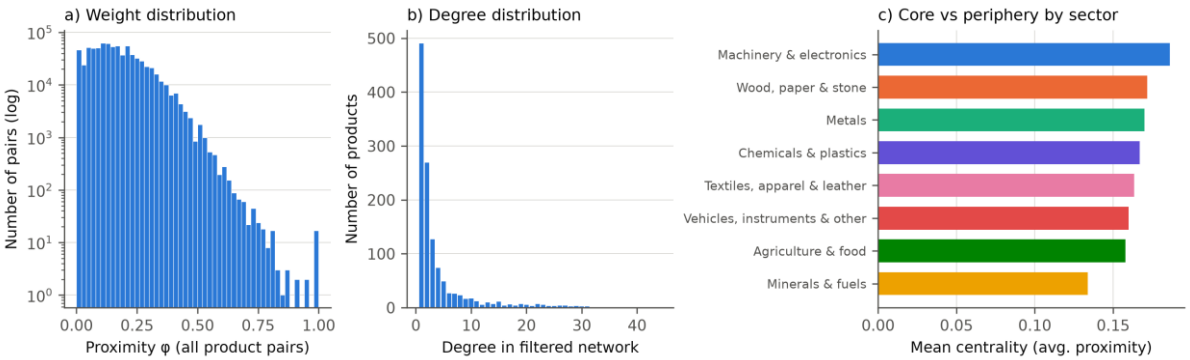
2. Describing the network

Weights: Most products are only weakly related to each other. The average proximity is 0.17, and only 0.4% of product pairs are above 0.5 (Figure 2a). A few pairs are very strongly related. 17 pairs have a proximity of exactly 1, for example four tin products (HS 8004–8007) and toys (9501–9503). These products have exactly the same exporters, so they are probably one industry split into several product codes.

Degree: Most products have very few links. The typical product has 2 links, and 40% of products have only one (Figure 2b). A few products are hubs with many links: articles of iron or steel (HS 7326) have 44 links, machinery for heating and cooling (8419) has 38, and farm machinery (8432) has 34.

Communities: The Louvain algorithm splits the network into 40 clear groups (modularity Q = 0.78). These groups often don't match the official sectors (NMI = 0.20). The largest group (188 products) puts cars, car parts and trucks together with electrical equipment. Other clear groups are:

- garments and shoes
- electronics (chips, computers)
- medical products (vaccines, medical instruments, also aircraft)
- raw materials (oil, gas, gold and food)



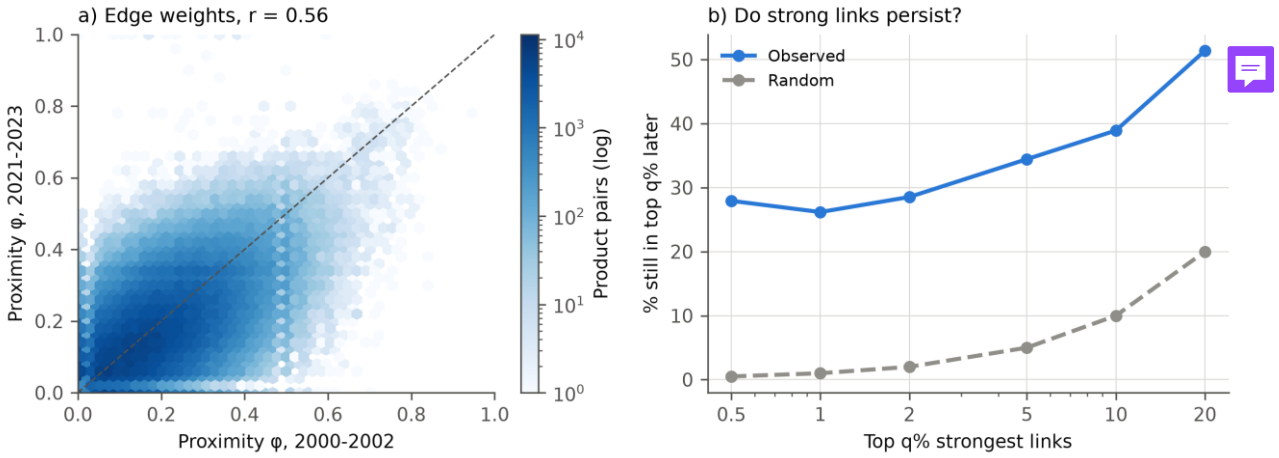
2. Figure

Core and periphery: I measured how central each product is by its average proximity to all other products. Machinery & electronics is the most central sector, and minerals & fuels is the least

central (Figure 2c). The most central products are machines and metal parts, like engines and motor parts. The most peripheral ones are raw materials, like cocoa waste, aluminium ores and asbestos, which have only one or two links each. More complex products tend to be more central (correlation with PCI: r = 0.27). This matches Hidalgo et al. (2007): complex goods are in the core and raw materials are at the edge. Garments are an exception. They form the densest group in the network, but they are not central to the network as a whole.

3. Comparing 2000-2002 and 2021-2023

I built the same network for 2000–2002 with the same 1,230 products (116 countries met the $1 billion threshold then). The links are clearly similar in the two periods (correlation r = 0.56, Figure 3a). Strong links usually stay strong: 28% of the strongest 0.5% of links in 2000–2002 are still among the strongest 0.5% twenty years later. By chance, this would be only 0.5% (Figure 3b). Products also stay about as central as they were (r = 0.68). In both periods machinery & electronics is the most central sector and minerals & fuels the least central.



3. Figure

Single links change more. Only 22% of the links in the 2000–2002 network are still in the 2021–2023 network, and the groups (communities) are only partly the same (NMI = 0.32). So the overall pattern of relatedness stays stable over twenty years, but individual links change a lot, so we should be careful when interpreting single links.

Limitations: Export values are not adjusted for inflation, so fewer countries reached $1 billion in 2000–2002. This means the two networks are based on different groups of countries. Some proximities are too high because one industry is split into several product codes. The communities also depend on how the network was filtered and which algorithm was used.